

# Colin Curtis - AI Systems and Software Engineering

 colinkeurtis |  Colin Curtis

## PROFESSIONAL SUMMARY

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Software engineering in the AI era is a complex and fast-moving profession where team-work and clear communication is as important as technical skill. Delivering great customer experience and value is the goal. In order to consistently reach that goal, cleanly-architected systems and a focus on cost-effective practices are the vehicle.

Currently a core contributor to the centralized LLM platform at a Fortune 100 financial services firm, my team's infrastructure serves 10,000+ users, hundreds of data scientists, and thousands of engineers.

Looking forward, I am actively studying and building applied agentic AI systems — combining ML modeling, RAG, and LLM orchestration — to deliver intelligent, cost-effective solutions to real business problems.

## EDUCATION

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2012 - 2014 M.S. (Physics) at **North Carolina State University**

2007 - 2012 B.S. (Physics w/ Math Minor) at **Appalachian State University**

## SKILLS AND INTERESTS

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- Professional
  - C4 software system design
  - SOLID, YAGNI, DRY coding principles
  - AI pair programming
  - Modeling, Linear Algebra, Calculus, Statistics and Error Analysis
- Personal
  - Gardening and Cooking
  - Sci-fi and Fantasy
  - Mountain Biking, Skiing, Running and Soccer

## APPLIED AI ENGINEERING — INDEPENDENT WORK

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### Independent Study

2026 – Present

- Agentic systems architecture: tool-use loops, multi-agent orchestration, human-in-the-loop patterns (sync gate, draft-queue, risk-tiered routing) and agent isolation via tool-router enforcement (IAM analogue)
- RAG architecture: chunking strategies, embedding model consistency, vector store selection and trade-offs (ChromaDB, pgvector, Pinecone), retrieval failure modes (corpus gap vs. query-form mismatch)
- Eval frameworks: heuristic gates, LLM-as-judge, RAGAS-style metrics (Precision@k, Recall@k, Faithfulness, Answer Relevancy) and golden test sets
- MCP server/client architecture, tool/resource/prompt primitives, team-wide tooling via `.mcp.json` and context-bloat management via the router pattern (N tools → 1 interface)
- Cost optimization: prompt caching, model tiering (Haiku/Sonnet/Opus by task complexity), batch processing, token budget design, and the naive RAG vs. tool-use cost/quality tradeoff with real token-cost math

- Designed end-to-end churn prediction system: time-boxed label definition (prediction window leads the renewal event), feature engineering across support, billing, and call data with explicit handling of NULL semantics, class imbalance (~5% positives after proper framing), and post-cutoff data leakage, wrote the full feature-extraction SQL from scratch
- Built a three-tier risk classifier (green/yellow/red) driving a RAG-backed agentic email system; retrieval grounds outreach in account-specific facts drawn from the CS playbook and interaction history
- Designed multi-layer eval framework: heuristic gates (block financial commitments, enforce 1:1 fact matching), LLM-as-judge (tone, grounding, content scope), and hand-crafted golden evals authored with domain experts; defined “false claim rate” as the primary production metric
- Diagnosed two independent production incidents from monitoring data alone: a feature feed failure (null rate stepping from 3% to 55%) and a silent LLM model swap; proposed remediation including emit-no-score-on-degraded-data, version pinning, and eval-gated model promotion
- Proposed treatment/control groups experimental design on at-risk accounts to measure statistically significant retention impact — named this as the only rigorous way to know whether the system actually works

# WORK EXPERIENCE

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## Machine Learning Ops and Software Engineering - Fidelity Investments April 2023 - Present

- Enterprise Data and AI (EDAI), AI/ML Solutions Engineering Squad
- Cloud Systems and Tech Stack:
  - AWS: Bedrock, Sagemaker, Cloudwatch, ECR, S3, RDS, EKS, IAM, Cost Explorer, etc.
  - Azure: OpenAI (OAI) portal + OpenTofu (Terraform) model deployment/management
  - python 3: botocore, flask, fastapi, pydantic, openai, streamlit, pandas, numpy, sqlalchemy, etc.
  - DevOps: Docker, kubernetes, Splunk, Argo, Jenkins, Snowflake, Datadog, jupyter NB, Open-Telemetry, Bruno
- Projects and Responsibilities:
  - Lead reviewer across squad's primary repositories for 3 years (the 2 repos representing 90% of team work), ranking #1 or #2 in review volume, code suggestions, and discussion depth across 10+ GitHub activity metrics - patterns established through review have become the squad's de facto engineering standards
  - Mentor engineers from junior through principal level across the squad and increasingly pulled into adjacent squads for expertise in MLOps, IaC, and LLM infrastructure
  - Onboarded GenAI Gateway to Fidelity's enterprise-wide API platform as one of only 8 core contributors, enabling secure LLM access for 10,000+ users; authored documentation supporting both internal teams and external consumers
  - Drove ML training/inference cost-savings initiative delivering five-figure monthly savings by implementing processes to detect and eliminate stagnant SageMaker deployments
  - Designed and shipped end-to-end automated testing pipeline for ML model deployment on AWS SageMaker, orchestrating parallel build/deploy/test/destroy workflows across Flask, DJL, and Triton model servers via Jenkins; supports hundreds of actively deployed models and up to 100 deployments per day. Resolved deep GPU/CUDA compatibility issues spanning NVIDIA drivers and SageMaker AMI configuration
  - Modularized MLOps model server build pipeline (Triton, DJL, custom) using Docker multi-stage builds, cutting repeat-build times by 10-60% per job and accelerating developer velocity for ~150 engineers consuming the pipeline
  - Shipped enterprise JWT authentication enabling per-user identity, request attribution, and fine-grained access control for data science workflows, while maintaining backward compatibility with existing OAuth2 flows; serves 1,000+ client integrations and growing steadily
  - Own Infrastructure-as-Code (OpenTofu/Terraform) for AWS Bedrock and Azure OpenAI deployments, managing dozens of customer accounts containing dozens of model deployments each, this system is onboarding ~2-3 new accounts per week and accelerating
  - Eliminated heavyweight CSP SDK dependencies (boto3, etc.) from core codebases by implementing pure HTTP request signing and composition logic, reducing dependency footprint and resolving cascading version conflicts that had bogged down build and runtime performance
  - Rebuilt Jenkins automation for GenAI Gateway integration and performance tests; added weekly scheduled runs alongside manual execution and extended suites to auto-upgrade lower-environment deployments to latest versions before testing — catching regressions earlier with no manual intervention
  - Founding collaborator on an agentic AI coding CLI — analogous to Claude Code — that performs repo-scouting, generates plans, and applies in-IDE edits by routing through the GenAI Gateway; served as primary technical advisor to the data-scientist stakeholder on Gateway capabilities and constraints, and drove sustained feature ideation, usability feedback, and testing contributions across the tool's multi-year development

## Senior Software Engineer - Garner Health

August 2022 - February 2023

- Systems and Tech:
  - Python 3: connexion/flask, psycpg2, sqlalchemy, boto3, pytest, alembic, asyncio, behave
  - AWS: S3, VPC, IAM, RDS, ECR, EKS, Step Functions, Lambda, Transfer Family, Secrets Manager
  - Docker, Kubernetes, Terraform
- Projects and Responsibilities:
  - PGP decryption module for the file ingestion system
  - Designed and implemented Client Data Configuration Versioning
  - System testing of RESTful APIs using Behavior Driven Design (behave library)
  - Product Requirement Documents for engineering/management communications

## Software Engineer - Actalent Services

April 2019 - July 2022

- Systems and Tech:
  - Microservices architecture for Bridgestone's tire-design CAD web application
  - Python 3, react.js, numpy, logging, pytest, mongoDB
- Projects and Responsibilities:
  - Designed and built a Dynamometer Dashboard for engine test data analysis at Ford Motors
  - Contributed to Bridgestone tire design CAD/analysis system written in Python 3 and React.js
  - Added 'overlays' to the CAD system, allowing tire engineers to view a transparency of one tire design overlaid on to another
  - Refactored the automatic Excel report generation - tire engineers click a button in the CAD system and the Python back-end generates a downloadable report
  - Converted and upgraded FMAT, a tire image analysis tool, from MATLAB to Python 3 and introduced 'alpha-shapes' computational geometry technique to improve the boundary estimates of complex 2-D shapes

## Junior Software Engineer - Renaissance Computing Institute

June 2018 - March 2019

- NIH National Center for Advancing Translational Sciences (NCATS) - Biomedical Data Translator
- Wrote and deployed APIs using Marathon, Mezos, Github, Jenkins, Docker, nginx, and python3

## Research Assistant, NCSU Physics

May 2013 - April 2018

- Designed and coded ALAI, a MATLAB application for automating fractal analysis of nanoscopic images
  - Reduced the user's active analysis time, per image, by a factor of ~50
- ANSYS Maxwell mesh-calculation to simulate 3-D electromagnetic fields
- LabVIEW software systems for instrument control and data collection
- Author/co-author of 3 peer-reviewed papers in experimental surface physics